

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

Code of Conduct:

*I **C.BHUVANESH (192521108)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: c.bhuvanesh

Annexure A

Comparative Analysis

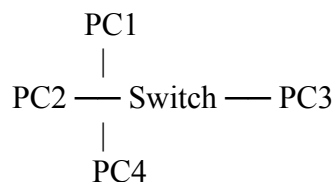
1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

The **star topology** and **bus topology** differ mainly in cost, reliability, and performance.

Comparison

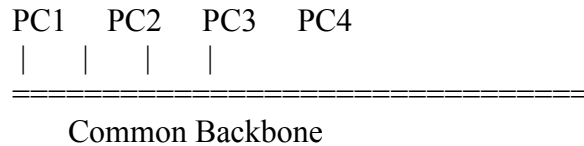
Feature	Star Topology	Bus Topology
Cost	Higher	Lower
Cabling	More cables required	One main backbone cable
Central device	Switch/hub required	Not required
Reliability	High	Low
Failure of one node	Usually affects only that node	Usually does not affect others
Backbone failure	Not applicable in the same way	Can affect the entire network
Performance	Better	Decreases as traffic increases
Collisions	Lower with switched Ethernet	Higher
Troubleshooting	Easy	More difficult
Scalability	Easy to expand	Limited

Star Topology



Each device has a separate connection to the central switch. If **PC2's cable fails**, the other computers can continue working.

Bus Topology



All devices share the same communication medium. If the **main backbone fails**, communication across the network can be disrupted.

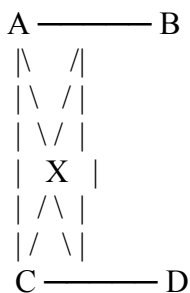
Conclusion

Bus topology is cheaper because it requires less cabling and no central switch. However, **star topology provides better reliability, performance, troubleshooting, and scalability**. Therefore, star topology is generally preferred for modern networks despite its higher initial cost.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

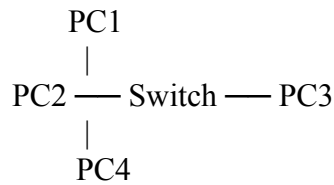
Mesh topology provides higher fault tolerance than **star topology** because it has multiple paths between devices.

Mesh Topology



- Devices have multiple connections.
- If one link fails, data can use another path.
- There is no single central point whose failure necessarily stops the network.
- It is highly reliable and suitable for **banks, data centers, and critical networks**.
- However, it is expensive and complex to install.

Star Topology



- Each device connects to a central switch.
- If one computer or cable fails, only that device is affected.
- However, if the **central switch fails**, all connected devices lose network connectivity.
- It is cheaper, easier to manage, and widely used in offices.

Comparison

Feature	Mesh	Star
Fault tolerance	Very High	Moderate
Alternative paths	Multiple	Usually limited
Central point of failure	No single central point	Central switch
Reliability	Very High	High
Cost	High	Lower
Complexity	High	Low

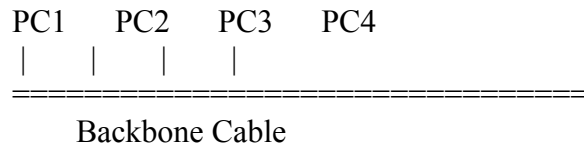
Conclusion

Mesh topology is more fault-tolerant and reliable because multiple alternative paths are available. **Star topology is simpler and less expensive**, but its central switch can become a single point of failure. Therefore, mesh is preferred for critical networks where maximum availability is required.

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

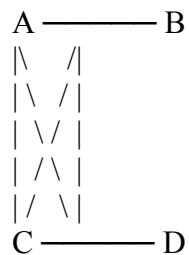
Bus topology is much simpler and cheaper to install than **mesh topology**.

Bus Topology



- Uses a **single backbone cable**.
- Requires fewer cables and connections.
- Easy and quick to install.
- Suitable for small, low-cost networks.
- Troubleshooting the backbone can be difficult.

Mesh Topology



- Each device has multiple connections to other devices.
- Requires many cables, ports, and network interfaces.
- Installation becomes increasingly complex as more devices are added.
- Provides excellent reliability and fault tolerance.
- More expensive and time-consuming to implement.

Comparison

Feature	Bus	Mesh
Installation	Simple	Complex
Cabling	Low	Very High
Cost	Low	High
Scalability	Limited	Complex
Maintenance	Relatively simple	Difficult
Fault tolerance	Low	Very High

For **n devices**, a full mesh requires:

$$2n(n-1)$$

links. For example, **10 devices require 45 links**, showing why mesh becomes expensive as the network grows.

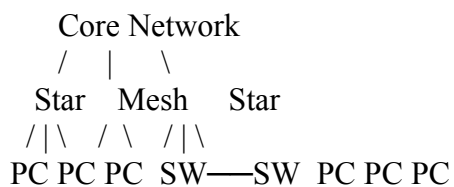
Conclusion

Bus topology is easier, cheaper, and faster to install, while **mesh topology requires more cables, ports, planning, and maintenance**. However, the extra complexity of mesh is justified in environments where **high reliability and fault tolerance** are more important than cost.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

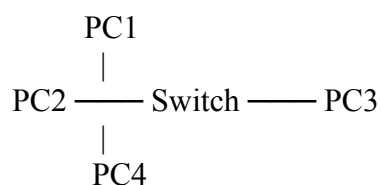
Hybrid topology generally provides greater scalability than star topology because it combines multiple topology types and allows different parts of a network to be expanded independently.

Hybrid Topology



- Combines two or more topologies such as **star, mesh, or bus**.
- New departments, buildings, or network segments can be added without redesigning the entire network.
- Different sections can use different topologies based on their requirements.
- Suitable for **large organizations, universities, and enterprise networks**.

Star Topology



- All devices connect to a **central switch or hub**.
- Easy to expand by connecting additional devices.
- Expansion is limited by the available ports on the central device.

- A larger switch or additional switches may be required when the network grows.

Comparison

Feature	Hybrid Topology	Star Topology
Scalability	Very High	High
Expansion	Flexible	Depends on switch capacity
Structure	Multiple topologies	Single topology
Cost	Higher	Lower
Complexity	Higher	Lower
Suitable for	Large networks	Small/medium networks

Conclusion

Hybrid topology offers greater scalability because different network sections can be expanded independently using the most suitable topology. Star topology is also easy to expand, but its growth depends more directly on the capacity and design of the central switching infrastructure. Therefore, **hybrid topology is better for large and continuously growing organizations.**

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

In terms of **maintenance and troubleshooting**, star topology is generally the easiest to manage, while mesh is the most complex.

Comparison

Topology	Maintenance	Reason
Star	Easy	Faults can be isolated to a particular device or cable
Bus	Difficult	Backbone failure can affect the entire network
Mesh	Very difficult	Large number of connections must be monitored

Hybrid Moderate Different topology types require different maintenance methods

1. Star Topology

- Easy to troubleshoot.
- If one cable fails, usually only one device is affected.
- Network administrators can quickly identify the faulty connection.
- Central switch makes management easier.

2. Bus Topology

- Troubleshooting is more difficult.
- A problem in the **backbone cable** can affect many or all devices.
- Locating the exact fault can take considerable time.

3. Mesh Topology

- Very reliable because of multiple paths.
- However, it has many cables and connections.
- Finding and maintaining a faulty link can be time-consuming.
- Requires skilled network administrators.

4. Hybrid Topology

- Combines multiple topology types.
- Maintenance complexity depends on the topologies used.
- Provides flexibility but requires careful monitoring and management.

Conclusion

Star topology is the easiest to maintain, while **mesh topology is the most difficult** because of its large number of connections. Bus topology is difficult to troubleshoot because of its shared backbone, while hybrid topology has **moderate complexity** due to its combination of different network designs.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

